

Geometry: Solving Parallelograms Using Properties



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DIRECTIONS

In parallelogram ABCD, E is the intersection of the diagonals. Use parallelogram properties to solve for x and find missing measurements. Show all work.

- 1 Find x and all side lengths. Perimeter = 26:

$$AB = 2x + 3, \quad BC = x + 1$$

Answer: _____

- 2 Consecutive Angles — Find x and both angle measures:

$$\angle A = (2x + 10)^\circ, \quad \angle B = (3x - 10)^\circ$$

Answer: _____

- 3 Diagonals Bisect — Find x and AC:

$$AE = 2x + 1, \quad CE = 3x - 4$$

Answer: _____

- 4 Find x and the perimeter. BC = AD = 7:

$$AB = 3x - 2, \quad CD = x + 10$$

Answer: _____

- 5 Find $m\angle B$, $m\angle C$, and $m\angle D$:

$$m\angle A = 65^\circ$$

Answer: _____

- 6 Diagonals Bisect — Find x and BD:

$$BE = 4x + 3, \quad DE = 6x - 5$$

Answer: _____

- 7 Opposite Angles — Find x and all four angles:

$$\angle A = (4x - 8)^\circ, \quad \angle C = (2x + 16)^\circ$$

Answer: _____

- 8 Opposite Sides — Solve for x and y:

$$AB = 2x + y = 11, \quad BC = x + 2y = 7$$

Answer: _____

- 9 Diagonals Bisect — Find x and AC:

$$AE = 3x - 1, \quad EC = x + 9$$

Answer: _____

- 10 Consecutive Angles — Solve for x and find both angles:

$$\angle C = (5x + 3)^\circ, \quad \angle D = (3x + 17)^\circ$$

Answer: _____

- 11 Find x and both side lengths. Perimeter = 50:

$$AB = 2x + 3, \quad BC = x + 7$$

Answer: _____

- 12 Diagonals Bisect — Find x, y, AC, and BD:

$$AE = 2x + 3, \quad CE = 4x - 5, \quad BE = y + 8, \quad DE = 3y - 4$$

Answer: _____



Answer Key & Solutions

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TEACHER NOTES

P1: Opp. sides equal. P2: Opp. angles equal. P3: Consecutive angles sum to 180° . P4: Diagonals bisect each other ($AE=CE$, $BE=DE$). Perimeter = $2(AB+BC)$.

1 Find x and all side lengths. Perimeter = 26:

$$AB = 2x + 3, \quad BC = x + 1$$

$$= x = 3, \quad AB = CD = 9, \quad BC = AD = 4$$

$$2(AB+BC)=26 \rightarrow AB+BC=13. (2x+3)+(x+1)=13 \rightarrow 3x=9 \rightarrow x=3.$$

2 Consecutive Angles — Find x and both angle measures:

$$\angle A = (2x + 10)^\circ, \quad \angle B = (3x - 10)^\circ$$

$$= x = 36, \quad \angle A = 82^\circ, \quad \angle B = 98^\circ$$

$$\angle A + \angle B = 180^\circ: (2x+10)+(3x-10)=180 \rightarrow 5x=180 \rightarrow x=36.$$

3 Diagonals Bisect — Find x and AC :

$$AE = 2x + 1, \quad CE = 3x - 4$$

$$= x = 5, \quad AC = 22$$

$$AE=CE: 2x+1=3x-4 \rightarrow x=5. AE=11, \text{ so } AC=2(11)=22.$$

4 Find x and the perimeter. $BC = AD = 7$:

$$AB = 3x - 2, \quad CD = x + 10$$

$$= x = 6, \quad P = 46$$

$$AB=CD: 3x-2=x+10 \rightarrow x=6, AB=16. P=2(16+7)=46.$$

5 Find $m\angle B$, $m\angle C$, and $m\angle D$:

$$m\angle A = 65^\circ$$

$$= m\angle B = 115^\circ, \quad m\angle C = 65^\circ, \quad m\angle D = 115^\circ$$

$$\angle C = \angle A = 65^\circ \text{ (opposite)}. \angle B = \angle D = 180^\circ - 65^\circ = 115^\circ \text{ (consecutive)}.$$

6 Diagonals Bisect — Find x and BD :

$$BE = 4x + 3, \quad DE = 6x - 5$$

$$= x = 4, \quad BD = 38$$

$$BE=DE: 4x+3=6x-5 \rightarrow 8=2x \rightarrow x=4. BE=19, \text{ so } BD=2(19)=38.$$

7 Opposite Angles — Find x and all four angles:

$$\angle A = (4x - 8)^\circ, \quad \angle C = (2x + 16)^\circ$$

$$= x = 12, \quad \angle A = \angle C = 40^\circ, \quad \angle B = \angle D = 140^\circ$$

$$\angle A = \angle C: 4x-8=2x+16 \rightarrow x=12. \angle A=40^\circ. \angle B=180^\circ-40^\circ=140^\circ.$$

8 Opposite Sides — Solve for x and y :

$$AB = 2x + y = 11, \quad BC = x + 2y = 7$$

$$= x = 5, \quad y = 1$$

$$y=11-2x; \text{ sub: } x+2(11-2x)=7 \rightarrow -3x=-15 \rightarrow x=5, y=1.$$

9 Diagonals Bisect — Find x and AC :

$$AE = 3x - 1, \quad EC = x + 9$$

$$= x = 5, \quad AC = 28$$

$$AE=EC: 3x-1=x+9 \rightarrow 2x=10 \rightarrow x=5. AE=14, \text{ so } AC=2(14)=28.$$

10 Consecutive Angles — Solve for x and find both angles:

$$\angle C = (5x + 3)^\circ, \quad \angle D = (3x + 17)^\circ$$

$$= x = 20, \quad \angle C = 103^\circ, \quad \angle D = 77^\circ$$

$$\angle C + \angle D = 180^\circ: (5x+3)+(3x+17)=180 \rightarrow 8x+20=180 \rightarrow x=20.$$

11 Find x and both side lengths. Perimeter = 50:

$$AB = 2x + 3, \quad BC = x + 7$$

$$= x = 5, \quad AB = CD = 13, \quad BC = AD = 12$$

$$2(AB+BC)=50 \rightarrow AB+BC=25. (2x+3)+(x+7)=25 \rightarrow 3x=15 \rightarrow x=5.$$

12 Diagonals Bisect — Find x , y , AC , and BD :

$$AE = 2x + 3, \quad CE = 4x - 5, \quad BE = y + 8, \quad DE = 3y - 4$$

$$= x = 4, \quad y = 6, \quad AC = 22, \quad BD = 28$$

$$AE=CE \rightarrow x=4, AE=11, AC=22. BE=DE \rightarrow y=6, BE=14, BD=28.$$

